```
# Install required libraries
!pip install transformers
!pip install torch

# Import libraries
from transformers import pipeline, AutoModelForSeq2SeqLM, AutoTokenizer
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7.1
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
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Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->tr
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=
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Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggi
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
```

```
# English → French
model_name_fr = "Helsinki-NLP/opus-mt-en-fr"
tokenizer_fr = AutoTokenizer.from_pretrained(model_name_fr)
model_fr = AutoModelForSeq2SeqLM.from_pretrained(model_name_fr)

def translate_en_to_fr(text):
    inputs = tokenizer_fr(text, return_tensors="pt")
    outputs = model_fr.generate(**inputs)
    return tokenizer_fr.decode(outputs[0], skip_special_tokens=True)

# English → German
model_name_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_de = AutoTokenizer.from_pretrained(model_name_de)
model_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_de)

def translate_en_to_de(text):
    inputs = tokenizer_de(text, return_tensors="pt")
    outputs = model_de.generate(**inputs)
    return tokenizer_de.decode(outputs[0], skip_special_tokens=True)
```

```
~/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip inst
arnings.warn("Recommended: pip install sacremoses.")
ding weights: 100%                                                    258/258 [00:00<00:00, 304.50it/s, Materializing param=model.shared.weight]
tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but
tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but
ig.json:        1.33k/? [00:00<00:00, 25.4kB/s]
nizer_config.json: 100%                                                 42.0/42.0 [00:00<00:00, 700B/s]
ce.spm: 100%                                                         768k/768k [00:00<00:00, 9.03MB/s]
et.spm: 100%                                                         797k/797k [00:00<00:00, 26.4MB/s]
ab.json:        1.27M/? [00:00<00:00, 10.2MB/s]
rch_model.bin: 100%                                                  298M/298M [00:03<00:00, 124MB/s]
lel.safetensors: 100%                                                298M/298M [00:03<00:00, 100MB/s]
ding weights: 100%                                                    258/258 [00:00<00:00, 310.60it/s, Materializing param=model.shared.weight]
tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but
tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but
eration_config.json: 100%                                               293/293 [00:00<00:00, 8.76kB/s]
```

```
sentences = [
    "I am learning Natural Language Processing.",
    "Machine learning is transforming the world.",
    "The exam will be conducted tomorrow.",
    "Can you help me with this assignment?",
    "Artificial Intelligence is the future of technology.",
    "The weather is very pleasant today.",
    "We are working on a translation project.",
    "This model converts English text into another language.",
    "Students must submit the assignment before deadline.",
    "Deep learning requires large amounts of data.",
    "How does a sequence to sequence model work?",
    "I completed my lab assignment successfully.",
    "The teacher explained the concept clearly.",
    "Programming becomes easy with regular practice.",
    "Natural language understanding is a difficult problem."
]
```

```
print("ENGLISH TO FRENCH TRANSLATION\n")

for sentence in sentences:
    translated = translate_en_to_fr(sentence)
    print("English :", sentence)
    print("French  :", translated)
    print()
```

```
ENGLISH TO FRENCH TRANSLATION

English : I am learning Natural Language Processing.
French  : J'apprends le traitement du langage naturel.

English : Machine learning is transforming the world.
French  : L'apprentissage automatique transforme le monde.

English : The exam will be conducted tomorrow.
French  : L'examen aura lieu demain.

English : Can you help me with this assignment?
French  : Pouvez-vous m'aider pour cette mission ?

English : Artificial Intelligence is the future of technology.
French  : L'intelligence artificielle est l'avenir de la technologie.

English : The weather is very pleasant today.
French  : Le temps est très agréable aujourd'hui.

English : We are working on a translation project.
French  : Nous travaillons sur un projet de traduction.

English : This model converts English text into another language.
French  : Ce modèle convertit le texte anglais en une autre langue.

English : Students must submit the assignment before deadline.
```

```
French  : Les étudiants doivent soumettre l'affectation avant la date limite.

English : Deep learning requires large amounts of data.
French  : L'apprentissage profond exige de grandes quantités de données.

English : How does a sequence to sequence model work?
French  : Comment fonctionne une séquence pour un modèle de séquence?

English : I completed my lab assignment successfully.
French  : J'ai terminé ma mission de labo avec succès.

English : The teacher explained the concept clearly.
French  : L'enseignant a expliqué clairement le concept.

English : Programming becomes easy with regular practice.
French  : La programmation devient facile avec une pratique régulière.

English : Natural language understanding is a difficult problem.
French  : La compréhension du langage naturel est un problème difficile.
```

```
print("ENGLISH TO GERMAN TRANSLATION\n")

for sentence in sentences:
    translated = translate_en_to_de(sentence)
    print("English :", sentence)
    print("German  :", translated)
    print()
```

```
ENGLISH TO GERMAN TRANSLATION

English : I am learning Natural Language Processing.
German  : Ich lerne Natural Language Processing.

English : Machine learning is transforming the world.
German  : Maschinelles Lernen verwandelt die Welt.

English : The exam will be conducted tomorrow.
German  : Die Prüfung wird morgen durchgeführt.

English : Can you help me with this assignment?
German  : Können Sie mir bei diesem Auftrag helfen?

English : Artificial Intelligence is the future of technology.
German  : Künstliche Intelligenz ist die Zukunft der Technologie.

English : The weather is very pleasant today.
German  : Das Wetter ist heute sehr angenehm.

English : We are working on a translation project.
German  : Wir arbeiten an einem Übersetzungsprojekt.

English : This model converts English text into another language.
German  : Dieses Modell konvertiert englischen Text in eine andere Sprache.

English : Students must submit the assignment before deadline.
German  : Die Studierenden müssen den Auftrag vor Ablauf der Frist einreichen.

English : Deep learning requires large amounts of data.
German  : Deep Learning erfordert große Datenmengen.

English : How does a sequence to sequence model work?
German  : Wie funktioniert eine Sequenz zum Sequenzmodell?

English : I completed my lab assignment successfully.
German  : Ich habe meinen Laborauftrag erfolgreich abgeschlossen.

English : The teacher explained the concept clearly.
German  : Der Lehrer erklärte das Konzept klar.

English : Programming becomes easy with regular practice.
German  : Programmierung wird einfach mit regelmäßiger Praxis.

English : Natural language understanding is a difficult problem.
German  : Natürliches Sprachverständnis ist ein schwieriges Problem.
```

```
complex_sentences = [
    "If the model is trained on large data, it will produce better results.",
    "Although the problem is difficult, we will try to solve it.",
    "Can you explain how machine translation works in simple words?",
    "Students who complete the assignment early will get bonus marks.",
```

```
    "Artificial intelligence is growing rapidly, and it is used in many applications."
]

for sentence in complex_sentences:
    translated = translate_en_to_fr(sentence)
    print("English :", sentence)
    print("French  :", translated)
    print()
```

```
English : If the model is trained on large data, it will produce better results.
French  : Si le modèle est formé sur de grandes données, il produira de meilleurs résultats.

English : Although the problem is difficult, we will try to solve it.
French  : Bien que le problème soit difficile, nous allons essayer de le résoudre.

English : Can you explain how machine translation works in simple words?
French  : Pouvez-vous expliquer comment la traduction automatique fonctionne en mots simples?

English : Students who complete the assignment early will get bonus marks.
French  : Les étudiants qui terminent l'affectation tôt obtiendront des points bonus.

English : Artificial intelligence is growing rapidly, and it is used in many applications.
French  : L'intelligence artificielle se développe rapidement et elle est utilisée dans de nombreuses applications.
```

Start coding or generate with AI.